Presentation and Management of Food Allergy in Breastfed Infants and Risks of Maternal Elimination Diets

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Breastfeeding is currently recommended as the optimal source of nutrition to infants. However, there are several studies that have shown clinical IgE- and non−IgE-mediated reactions to foods in exclusively breastfeeding infants, specifically to cow's milk, egg, peanut, and fish. Literature suggests that antigenic food proteins present in human milk can be found in substantial enough amounts to elicit clinical reactions in some, already-sensitized infants, including anaphylaxis, eczema exacerbation, and non−IgE-mediated gastrointestinal food-allergic syndromes. Diagnosis of food allergy in a breastfed infant and identification of the trigger foods in the mother's diet can be especially challenging in infants with delayed symptoms, such as eczema and gastrointestinal symptoms. Management is further complicated in infants with atopic dermatitis, who have increased caloric needs and therefore in whom nutrition is an extremely important factor for growth and development. One needs to balance possible benefits with risks of further food sensitization through the skin when foods are eliminated from their diets. We review here the literature on clinical presentation and evidence for food allergy in exclusively breastfed infants, including the presence of food antigens in human milk. Incorporating clinical experience and the available data, which largely come from case reports and small, nonrandomized studies performed in referral centers with several limitations, we propose a novel algorithm to diagnosis and management, with emphasis on nutritional considerations. **(J Allergy Clin Immunol Pract 2020;8:52-67)**

Key words: *Breastfeeding; Food allergy; Nutrition; Food antigen; Atopic dermatitis*

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INTRODUCTION

Breastfeeding is the optimal source of nutrition for infants; however, management of breastfeeding infants who present with signs and symptoms of food allergy can be challenging. Infants may present with atopic eczema, hives, and gastrointestinal symptoms that may be a manifestation of food allergy, but these symptoms can also appear without association with food allergies. A clinical conundrum encountered by physicians in daily management of such breastfed infants is whether a mother needs to avoid highly allergenic foods. The considerations in such cases include whether symptoms are truly attributable to food allergy and whether they are considered significant enough to necessitate maternal elimination diets. The decision to eliminate foods should be made by balancing the benefits with possible risks of the nutritional hazards of a limited diet on infants (and mothers). With unnecessary dietary restrictions, there may be implications for tolerance development, especially in infants with atopic eczema who are at increased risk for sensitization through the skin. There are no current guidelines about how to approach a breastfed infant in whom food allergy may be contributing to symptoms. We review here the literature on clinical presentation and evidence for food allergy in breastfed infants, including the presence of food antigens in human milk, and propose a novel algorithm to diagnosis and management, with emphasis on nutritional considerations.

PRESENTATION OF FOOD ALLERGY TO FOOD ANTIGENS IN HUMAN MILK IN BREASTFED INFANTS

Anaphylactic reactions

Anaphylactic reactions to human milk are rarely reported, and although no estimates of the prevalence of these reactions exist, we consider this to be a rare manifestation of food allergy in the exclusively breastfed infant. However, anaphylactic reactions can present differently in infants and symptoms can be nonspecific, which can make recognition of anaphylaxis more difficult.[1]

Two cases of immediate reactions to human milk after maternal ingestion of fish have been reported in exclusively breastfed infants.[2,3] An episode of anaphylaxis (urticaria, vomiting, cough, and wheeze) was described in a 4-month-old infant

Abbreviations used
CM- Cow's milk
CMA- Cow's milk allergy
EoE- Eosinophilic esophagitis
FPIAP- Food protein–induced allergic proctocolitis
FPIES- Food protein–induced enterocolitis syndrome
GER- Gastroesophageal reflux
SPT- Skin prick test

within minutes of breastfeeding, with maternal ingestion of fish hours earlier, without evidence of accidental direct exposure to fish proteins in the infant.[3] Specific IgE was positive to various fish.[3] Similarly, wheezing and urticaria within an hour of breastfeeding were described in a 5-month-old infant, linked to maternal ingestion of fish hours before. Skin prick test (SPT) result of the infant was positive to several fish species.[2] In both cases, avoidance of fish in the mother's diet caused resolution of symptoms, and subsequent exposure of fish via human milk[2] or direct ingestion[3] caused symptoms to recur.

Lifschitz et al[4] describe a 5-day-old breastfed infant who developed proctitis with frankly bloody stools who was switched to a casein-based formula on day 8 of life. After the third feed of casein-based formula, the infant became lethargic, vomited profusely, had bloody loose stools, and became hypotensive and cyanotic (timing of reaction was not described). The mother eliminated all cow's milk (CM) from her diet. At 3-weeks-old, the infant resumed breastfeeding without symptoms, but when accidentally fed previously frozen milk collected before the mother had eliminated CM, the infant developed cyanosis, undetectable blood pressure, and mucousy, bloody loose stools and required resuscitation. Casein hydrolysate formula was initially tolerated, but after 2 days, the infant had another episode of duskiness, floppiness, hypotension, and facial swelling and was treated with epinephrine. The infant had positive specific IgE to α-lactalbumin and β-lactoglobulin, and elevated peripheral blood eosinophils, but outgrew the allergy, tolerating direct ingestion of CM by age 12 months.[4] These rare case reports suggest that food proteins in the maternal diet can elicit anaphylactic symptoms in a breastfed infant.

Atopic dermatitis

The prevalence of eczema ranges from 10% to 30% in children in industrialized countries,[5,6] with about 16% of all children presenting in the first year of life.[6] Among infants with moderate to severe eczema, about one-third (28%-37%) are likely to have clinically diagnosed immediate-type IgE-mediated food allergy.[7-9] These trigger foods result in eczema flare-ups that typically occur between 6 and 48 hours after ingestion of the food.[10] The exact role of foods in provocation of atopic dermatitis flares is less clear and can be difficult to evaluate in a clinical setting.[11]

In infants, the role played by maternal dietary antigens in human milk, especially CM protein, in provoking atopic dermatitis flares is a common concern among caregivers[11]; however, causality is more difficult to infer in the clinical practice. In a population-based cohort of 1749 infants in Denmark, 39 (2.2%) developed cow's milk allergy (CMA); of these, 17 (44%) developed symptoms during breastfeeding. Among them, 9 were exclusively breastfed; of these, 8 responded to maternal elimination diet and developed atopic dermatitis after maternal consumption of a minimum of 0.5 L of CM daily.[12] Cant et al[13] investigated the impact of maternal dietary restriction followed by introduction of egg and CM in 37 breastfed infants with atopic dermatitis, with all but 5 exclusively breastfed. None were given CM or egg, and no new solids were introduced during the trial. In a series of 2 trials, 6 infants showed improvement on egg- and dairy-free diets and symptom recurrence with reintroduction of these foods into the maternal diet. This study suffered from methodological issues (soy, a common allergen, replaced egg and dairy-based formula for maternal consumption in the first trial); however, their results indicate that a proportion (16%) of infants in their sample may have had food-allergic reactions due to the maternal diet.[13] Our previous study evaluated responses to CM in the maternal diet in 27 breastfeeding infants, 17 with and 10 without CMA.[14] Of the 17 infants with CMA, 14 were exclusively breastfed. All mothers restricted CM and other common allergens for 2 to 4 weeks before consuming CM in advance of physician-supervised provision of breastfeeding in a clinic setting. All but 1 of the CM-allergic infants experienced eczematous reactions between 2 and 80 hours from the last dose. None of the infants without CMA demonstrated symptoms.[14] We believe that direct clinical observation of the infant before and after maternal CM ingestion provides a higher level of evidence for the role that CM played in the infant's AD exacerbations. The most recent report from Japan evaluated the role of tree nut–related foods (coffee and chocolate) and fermented foods (cheese, yogurt, bread, soy sauce, miso soup, and fermented soy beans) in maternal diet on 92 exclusively breastfed infants with atopic dermatitis and reported that 73% of the infants showed improvement with a maternal elimination diet of these foods.[15] Infants were then challenged at home to individual foods by maternal consumption for 3 consecutive days; all infants had an eczematous reaction by day 3 to at least 1 of the foods.[15] The common trigger foods were chocolate, yogurt, soy sauce, and miso soup.

In summary, in a limited number of studies, most of which are not based on general populations, resolution of symptoms with food avoidance followed by exacerbation with reintroduction of individual foods indicate a role for foods, especially CM, in causing atopic eczema through the maternal diet in a proportion of breastfed infants with CMA. However, the overall prevalence of CMA is low and most infants with eczema do not have CMA.

Food protein–induced allergic proctocolitis

Food protein–induced allergic proctocolitis (FPIAP) is common in breastfed infants, with up to 60% of cases reported among exclusively breastfed infants.[16] Foods are thought to be a trigger for many; however, concomitant viral infections may cause a similar picture with bloody stools. In FPIAP, large amounts of frank blood or infant colic are unusual. Greater than 20% of breastfed infants with FPIAP have concomitant atopic dermatitis, and 25% of affected infants have an atopic family member.[17]

In addition to CM, maternal intakes of soy, egg, corn, and wheat have been attributable foods in exclusively breastfed infants with FPIAP,[17,18] and reactions to more than 1 food are possible.[19-21] Erdem et al[22] reported a retrospective study of 77 patients with FPIAP, 41 (53%) of whom were breastfeeding at symptom onset, and found that nearly 80% of infants reacted to CM alone, 13% to both CM and egg, 5% to egg alone, and 4%

to multiple foods.[22] All infants who presented during exclusive breastfeeding had symptoms resolve with elimination of CM from the maternal diet.[22] In another cohort of 37 infants with FPIAP, 41% exclusively breastfed at symptom onset, all patients reacted to CM and 14% to egg white.[21] Similarly, Kaya et al[23] found that 100% of their cohort of 60 patients with FPIAP reacted to CM, and 17% had an additional food trigger.

A flexible sigmoidoscopic evaluation of 22 infants with rectal bleeding[24] found that 64% met diagnostic criteria for FPIAP after clinical and histologic evaluation. Mothers of exclusively breastfeeding infants were instructed to remove all sources of CM from their diet. In 5 of 6 infants, symptoms resolved between 2 and 8 weeks after the initiation of a CM-free diet; 1 infant required amino acid–based formula, suggesting that either the mother did not effectively remove CM completely from her diet or that CM may have not been the sole allergen responsible for symptoms.[24]

In summary, probably the best data on role of maternal ingestion of CM exist in FPIAP.

Food protein–induced enteropathy syndrome

In infants, chronic food protein–induced enteropathy syndrome (FPIES) typically presents as episodes of vomiting and chronic diarrhea with or without blood without a clear pattern in relation to the timing of meals. Over time, symptoms can progress, resulting in failure to thrive, lethargy, dehydration, and metabolic disturbances, and nutritional deficiencies can occur.[17,25] Contrastingly, acute FPIES presents with multiple (up to 10-20) episodes of vomiting (often projectile), pallor, dehydration, hypotension, and lethargy within 1 to 4 hours of ingestion of the food trigger, followed by metabolic disturbances, acute neutrophilia, leukocytosis, and other abnormalities that may mimic sepsis up to 6 hours after ingestion.[17,25] Chronic FPIES develops in the context of consistent exposure to the trigger food, and acute FPIES reactions occur when foods are ingested with less regularity or after a significant interval of avoidance (acute-on-chronic FPIES).[17,25] FPIES reactions are most commonly triggered by CM and soy, and solid foods are responsible for a smaller proportion of reactions.[26-28]

The first report of FPIES in an exclusively breastfed infant described both chronic and acute FPIES in a 1-month-old infant presenting with symptoms of chronic FPIES while the mother had an unrestricted diet. Symptoms resolved when CM was removed from the maternal diet with continued breastfeeding and supplementation of extensively hydrolyzed formula. Acute FPIES reactions were induced with CM-based formula weeks after symptom resolution when there was accidental maternal ingestion of dairy followed by a breastfeed.[29] The second report described acute FPIES triggered by soy after the infant breastfed after maternal ingestion of a large dose of soy-containing ice cream.[30] The third report documented 2 cases of acute FPIES induced by breastfeeding after maternal consumption of CM, each with induction of acute symptoms (repetitive vomiting, pallor, and diarrhea) after direct ingestion challenge of a CM-based formula by the infant (timing of symptoms not specified), and resolution after maternal dietary elimination of all CM products.[31] Most recently, among a cohort of 27 patients with FPIES, there were 6 cases (22%) of FPIES while exclusively breastfed, and a proportion of these cases showed improvement with restriction of the maternal diet alone.[21] Their reported rate of FPIES in breastfed infants is higher than the rates described by other authors; however, the details of these reactions and maternal dietary changes were not described in detail.

Cases of chronic FPIES induced by maternal dietary proteins in human milk have also emerged. Two cases described breastfed infants with regurgitation, colic, diarrhea, and failure to thrive; in both, subsequent acute episodes due to the maternal ingestion of CM were also reported.[32] Caubet et al[27] report that of 160 patients with FPIES, 3 infants (2%) had chronic FPIES symptoms while being exclusively breastfed, although specific trigger foods were not identified in this report. The authors suggest that given the relatively low concentration of food allergens in human milk, or as a result of other protective factors in human milk such as IgA antibodies and partially processed food proteins, acute FPIES reactions were not seen in these patients.

These few cases and case series suggest a role for dietary proteins excreted in human milk eliciting symptoms of both acute and chronic FPIES, although the prevalence is unknown.

Eosinophilic esophagitis

To our knowledge, there have been no reports to date of eosinophilic esophagitis (EoE) diagnosis specifically in breastfed infants. Given that the prevalence of this condition is relatively rare (0.5-1 case per 1000),[33] there is hesitation to conduct endoscopy for diagnosis in younger children and infants. Existing studies evaluating pediatric EoE populations do contain infants[34,35]; however, there is no information regarding specifics of the infant's diet at the time of symptom onset or the number of infants in these samples. Our clinical experience suggests that some patients with EoE have initial gastrointestinal symptoms during breastfeeding, which often resolve on initiation of hypoallergenic formula, which is widely available, and present only when highly allergenic foods such as CM are reintroduced.

A summary of reactions in breastfed infants is provided in Table I.

FOOD ANTIGENS IN HUMAN MILK

Dietary antigens can be detected in mothers' milk, including ovalbumin,[36,37] β-lactoglobulin,[14,38] gliadin,[39] and peanut,[40-43] but the measured concentrations are often in the low nanogram per milliliter range and the clinical significance is often unknown. Across the studies, about one-half to two-thirds of mothers appear to secrete dietary food proteins in their milk, and among those who do, levels vary and not all infants develop symptoms.[36-38,40,41,43] Several reports have found large interindividual variation in food allergens in human milk and have demonstrated that up to half the women had undetectable levels of food allergens after the consumption of these foods. After their ingestion, the detection of specific foods, including CM, peanut, and egg, varies greatly from subject to subject in terms of the timing of the onset of detection (1-2 hours for CM, 1-8 hours for peanut, and up to 8 hours for egg), the peak concentration (up to 10 ng/mL for β-lactoglobulin, 300 ng/mL for peanut, 50-400 ng/mL for Ara h 2 and Ara h 6, 4-8 ng/mL for hen's egg ovalbumin with even occasionally higher levels reported), and the duration throughout which the food protein is detectable (1-12 hours).[14,36-38,41-44] After CM consumption, 15% to 47% of subjects had no detectable β-lactoglobulin in their milk.[14,38] More than 25% of mothers in egg consumption arms of a randomized trial demonstrated no detectable ovalbumin in their milk,[36] and in a randomized cross-over trial, 24% of mothers never demonstrated detectable ovalbumin in their milk.[37] After

TABLE I. Summary of IgE- and non−IgE-mediated clinical reactions of food allergy in breastfed infants

Author, year	Country	Study design	Study n	Trigger foods	Infant age	Infant feeding method	Study description	Results
Anaphylactic reactions to breast milk								
Monti et al,[3] 2006	Italy	Case report	1	Fish	4 mo	Exclusively breastfed	Observational only, no controlled challenge	Urticaria, vomiting, wheeze within minutes of initiation of breastfeeding ∼3 h after maternal ingestion of fish. Child had positive IgE to 2 fish species
Arima et al,[2] 2016	Japan	Case report	1	Fish	5 mo	Exclusively breastfed	Observational only, no controlled challenge	Urticaria ∼1 h after breastfeeding after maternal fish consumption. Infant demonstrated positive SPT result to several fish species
Lifschitz et al,[4] 1988	USA	Case report	1	CM	5 d	Casein-based formula, followed by total breastmilk feeding	Observational only, no controlled challenge	Infant with grossly bloody stools while breastfeeding, switched to a casein-based formula, developed lethargy, profuse vomiting, bloody stools, hypotension, hypoperfusion, cyanosis. Infant positive RAST result for α- lactalbumin and β-lactoglobulin
AD								
Host et al,[12] 1988	Denmark	Subset of a prospective cohort study	9	CM	<4 wk-12 wk	Exclusively breastfed, all supplemented with casein-based formula	Allergist- confirmed CMA (either complete resolution of symptoms with maternal restriction of CM or observed direct oral challenge of the infant)	On infant direct challenge to human milk after maternal CM ingestion, 8 of 9 infants demonstrated AD reactions, additional symptoms including vomiting, diarrhea, colic, wheeze, and rhinorrhea; 5 infants were found to have IgE-mediated CMA defined by SPT and RAST
Cant et al,[13] 1986	England	Double- blind crossover trial	37	CM, egg	6 wk-6 mo	Breastfed (exclusivity not specified)	Maternal dietary restriction followed by maternal ingestion and breastfeed in infants with AD	6 of 37 infants' AD improved with CM/egg with exclusion and worsened with reintroduction

(continued)

TABLE I. *(Continued)*

Author, year	Country	Study design	Study n	Trigger foods	Infant age	Infant feeding method	Study description	Results
Uenishi et al,[15] 2011	Japan	Prospective research study	92	Tree nut—related foods, fermented foods	3-8 mo	Breastfed (exclusivity not specified)	Elimination diet in mother for 2 wk followed by reintroduction	Soy sauce and chocolate were the most common trigger foods for AD, although all foods tested were shown to trigger reactions in multiple infants. Average number of food triggers identified was 1.6 (range, 1-4); 8 infants had exacerbation upon rechallenge
Jarvinen et al,[14] 1999	Finland	Prospective research study	17 infants with CMA, 10 healthy infants	CM	1.8-9.4 mo	Breastfed (some exclusive, others with formula)	Restricted maternal diets 2-4 wk, challenge with maternal ingestion via breastfeeding in infants. Infants showing mild or no symptoms were challenged with direct consumption of CM formula	16 of 17 infants with CMA demonstrated cutaneous reactions to CM through human milk, some also having GI symptoms or upper airway symptoms (eg, rhinorrhea). Control infants showed no symptoms. SPT result to CM was negative in all but 1 infant with CMA, and no infants showed positive SPT result to their mother's milk
FPIAP								
Erdem et al,[22] 2017	Turkey	Retrospective cross-sectional study	77	Multiple food triggers identified	Mean age, 3.31 mo (range, 1 wk-36 mo)	41 of 77 being breastfed at evaluation (exclusivity not specified)	Elimination diet in mother, evaluated for symptom resolution within 72-96 h, and trial reintroduction of the food	Among all patients, 78% reacted to CM only, 13% to both CM and egg, 5% to egg alone, and 4% to multiple foods. All infants who presented during exclusive breastfeeding had symptoms resolve with elimination of CM from the maternal diet
Arik Yilmaz et al,[21] 2017	Turkey	Prospective research study	37	Multiple food triggers identified	Median, 2 mo (IQR, 1-3)	33 of 37 were breastfeeding at evaluation		Trigger food eliminated from maternal diet in 32 of 33 breastfed infants, all but 3 had resolution with maternal diet changes only. CM was a trigger in all infants, 5 infants were also triggered by egg white, and 8 others reacted to other foods (legumes, chestnut, soy, rice, or meats)

Kaya et al,[23] 2015	Turkey	Prospective research study	60	Multiple food triggers identified	Mean age, 1.7 mo (range, 1 wk-6 mo)	23 breastfeeding (exclusivity not specified)	Elimination diet, evaluated for symptom resolution within 72-96 h, and resumption of symptoms with reintroduction of the food	100% reacted to CM, only trigger in 83% (50) of infants. CM and egg were triggers for 4 infants, and CM and other foods in 4 patients (chicken, wheat, potato); triggers were not identified for 2 patients who had symptoms resolve with cessation of breastfeeding and consumption of amino acid −based formula
Xanthakos et al,[24] 2005	United States	Prospective cohort study	22	CM	<6 mo	12 of 22 were breastfed at enrollment	Breastfeeding mothers eliminated all CM from their diets, and sigmoidoscopy was repeated at 3 wk, and again at 6 wk if symptoms and histological features of colitis had not resolved at 3 wk	8 of 12 (66.7%) breastfeeding infants were diagnosed with FPIAP (60% of formula-feeding infants were diagnosed with FPIAP); 5 of these infants had complete follow- up (9 wk) and resolution of bleeding; average time to symptom resolution was 5.6 wk (range, 2-8 wk). Patients with FPIAP had higher total IgE blood levels, and higher serum eosinophil levels
FPIES								
Monti et al,[29] 2011	Italy	Case report	1	CM	1 mo	Exclusively breastfed	Observational only, no controlled challenge	Infant presented at age 1 mo with chronic FPIES with unrestricted maternal diet. Resolution of symptoms with CM-free maternal diet, and addition of extensively hydrolyzed formula. SPT result and IgE were negative
Tan et al,[30] 2012	Australia	Case report	1	Soy	5 mo	Exclusively breastfed	Observational only, no controlled challenge	Infant developed symptoms hours after feedings of soy-based formula. At 6 mo, infant developed acute FPIES symptoms hours after breastfeeding following maternal ingestion of a large amount of soy-based ice cream. SPT result to soy was negative

(continued)

TABLE I. *(Continued)*

Author, year	Country	Study design	Study n	Trigger foods	Infant age	Infant feeding method	Study description	Results
Kaya et al,[31] 2016	Turkey	Case series	2	CM	15 d, 2.5 mo	Exclusively breastfed	Elimination diet, direct oral challenge after symptom resolution, infants were tested using direct oral challenge, atopy patch testing, SPT, and IgE testing	Both infants had acute FPIES reactions after breastfeeding while mothers' diets included CM, each with induction of acute symptoms after direct ingestion by the infant, and resolution after maternal dietary elimination of all CM products. Atopy patch testing was positive for 1 infant, and CM SPT result and IgE were negative in both infants
Miceli Sopo et al,[32] 2014	Italy	Case series	2	CM	3 mo and 2 mo	Exclusively breastfed	Observational only, no controlled challenges	Infant 1: Chronic GI symptoms (colic, diarrhea, failure to thrive, pallor), which resolved after elimination of CM from the maternal diet, acute FPIES symptoms after accidental maternal ingestion of CM and subsequent breastfeeding. Infant 2: chronic symptoms (colic, regurgitations, diarrhea) with breastfeeding, and had projectile vomiting, diarrhea, pallor, and lethargy with the first 2 exposures to CM formula. SPT result was negative for both infants

Caubet et al,[27] 2014	United States	Retrospective and prospective cohort study	160	Multiple food triggers identified	Median age at diagnosis, 15 mo	3 infants developed symptoms while being exclusively breastfed	Direct OFCs were offered to patients no less than 12 mo after the last reaction to the suspected food	Of 160 cases of FPIES, only 3 infants (2%) exhibited chronic FPIES symptoms during exclusive breastfeeding, although specific trigger foods were not specified in this report. In the full cohort, most common foods were CM, soy, rice, oat, fish, and shellfish; 24% of the full cohort had elevated IgE to the food triggering food allergy, 39% had IgE sensitization to nontrigger foods
Arik Yilmaz et al,[21] 2017	Turkey	Prospective research study	27	Multiple food triggers identified	Median 4 mo	23 breastfeeding at evaluation	Retrospective review—evaluated infants with FPIES, mothers were instructed to eliminate suspected foods from diet, and OFC was conducted	15 of the 23 breastfed infants' causative foods were identified and eliminated from maternal diet, 13 of these infants' symptoms resolving with maternal dietary changes only. Causative foods in the full FPIES sample were CM in 74.1%, fish in 14.8%, egg white in 11.1%, egg yolk in 11.1%, and other foods in 1 child each (rice, potato, wheat, and banana); 22% of infants were triggered by more than 1 food

AD, Atopic dermatitis; *GI*, gastrointestinal; *IQR*, interquartile range; *OFC*, oral food challenge; *RAST*, radioallergosorbent test.

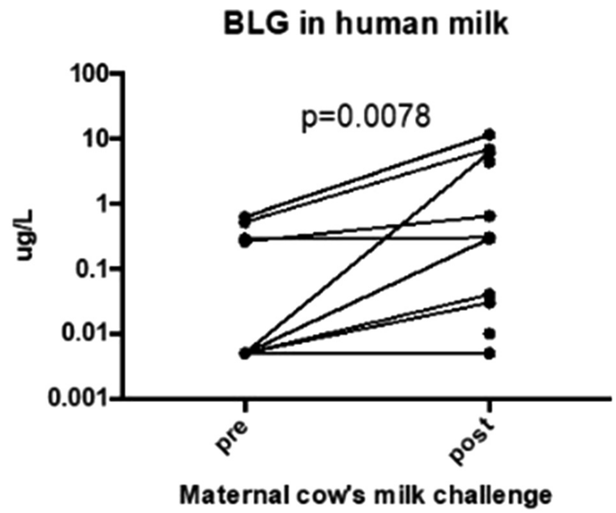

FIGURE 1. β-Lactoglobulin (BLG) levels in mothers' milk during a CM challenge on the mother. Samples assessed were collected after a 2-week strict milk elimination diet (pre) and after symptoms appeared during the challenge (post). Post noted at time of onset of symptoms. BLG levels measured by sensitive in-house ELISA. Modified from Jarvinen et al.[14]

peanut consumption, 52% of women had no detectable peanut proteins,[41] and 72% had no detectable Ara h 2, the most potent allergenic peanut protein, in human milk.[43] Overall, levels were highly variable, kinetics differ with each food, and the clinical importance of low positive levels is unclear; peanut may exist in immune complexes with IgA and IgG as more recently reported.[45] One study noted that mothers with an infant with CM allergy appear to have a different pattern of secretion of β-lactoglobulin with some "slow metabolizers" compared with mothers with healthy children.[38]

One possible reason for this variability is the difference in detection methods that are used as well as variations in sampling and possible food-specific kinetics. The sensitivity at which peptides resulting from gastric digestion are detected may have a major effect, and methods relying on the presence of an intact protein may not give an accurate picture of antigen transport to milk. Sensitive detection of ingested protein antigens can be achieved by measuring inhibition of antibody binding has recently been reported for Ara h 2 in human milk.[43]

Dietary antigen challenge through mother's milk

Skin prick testing using breast milk collected after maternal ingestion of CM, egg, and peanut has been shown to induce positive reactions in sensitized infants who experienced symptoms during breastfeeding.[46] The role for dietary antigens in human milk in inducing symptoms in already-sensitized infants was demonstrated by us during a physician-supervised oral food challenge that was performed on infants through human milk.[14] Twenty-seven mothers were strictly avoiding all dairy and CM protein in their diets for a minimum of 2 weeks before they arrived at the hospital for CM challenge with their breastfed infants, aged 1.8 to 9.4 months, all free of symptoms of eczema or other symptoms at the time of challenge. Increasing doses of CM up to 700 mL were ingested by the mothers on day 1, which occurred in the hospital; thereafter, free consumption was allowed from day 2 onward at home. Infants were breastfed at 1, 2, and 4 hours after beginning of the challenge and followed for 2 additional hours in the hospital. Eleven of those with positive CM challenge through human milk reacted in 2 to 9 hours from the last dose on day 1. β-Lactoglobulin levels measured by sensitive in-house ELISA were shown to increase significantly after antigen challenge[14] (Figure 1). The presence of other clinically relevant CM allergens, such as caseins, was not assessed.

Allergic reactions to endogenous human milk antigens

Occasionally, symptoms of food allergy do not resolve after extensive and strict elimination of foods in mother's diet. Although this is rare, the clinical experience is that these infants' symptoms resolve only after discontinuation of breastfeeding and initiation of hypoallergenic formula. Several possibilities exist: (1) the extent of or adherence to maternal food elimination diet is not enough, (2) symptoms are not related to food allergy, or (3) infant was reacting to endogenous human milk proteins. Regarding the latter, among the major allergenic proteins in CM, α-lactalbumin, α_{s1}-casein, β-casein, and k-casein are also found in human milk, with α-lactalbumin and β-casein being the most abundant.[47] The amino acid sequences of bovine and human α-lactalbumin are 78% identical, whereas β-casein and κ-casein are 49% and 52% identical.[48] The specific IgE-binding epitopes on the major CM allergens have extremely high identity with corresponding human milk proteins.[48] Using immunoblot inhibition, IgE reactive to human milk antigens was detected in up to 80% of CM-allergic subjects, some demonstrating positive skin test reactions to human milk.[49] Consequently, as we showed in our previously published report,[48] a high degree of similarity exists between the IgE-binding regions on bovine and human milk proteins. In the same publication, we showed the presence of IgE antibodies to human milk in up to 60% of milk-allergic infants, who were showing symptoms despite maternal strict and extensive food elimination diet.[48] More importantly, these patients' IgE showed IgE cross-linking with human milk α-lactabumin, using rat basophilic leukemia mediator release assay (RBL-2H3 cells). Two recent reports also showed recognition of human milk and human milk α-lactalbumin by the basophil activation assay using CM-allergic subjects' sera.[50,51] These data suggest that sensitization to mothers' milk, albeit rare, can occur.

DIAGNOSIS AND SUGGESTED WORKUP OF POSSIBLE FOOD ALLERGY IN A BREASTFED INFANT

Infant presenting with IgE-mediated symptoms

In the setting of a clear history of IgE-mediated symptoms immediately after breastfeeding, such as hives, vomiting, cough, or wheezing, workup of food allergy should be strongly considered. The best tool for identifying triggers is detailed history including symptoms (timing, severity) and their association with specific foods and symptom resolution upon maternal avoidance. This is best accomplished using a food and symptom diary, followed by consideration of SPT and/or IgE testing to suspected foods. As discussed above, there are rare reports of anaphylaxis in exclusively breastfed infants, specifically with maternal ingestion of CM and fish; however, other top allergens should be considered on the basis of history.

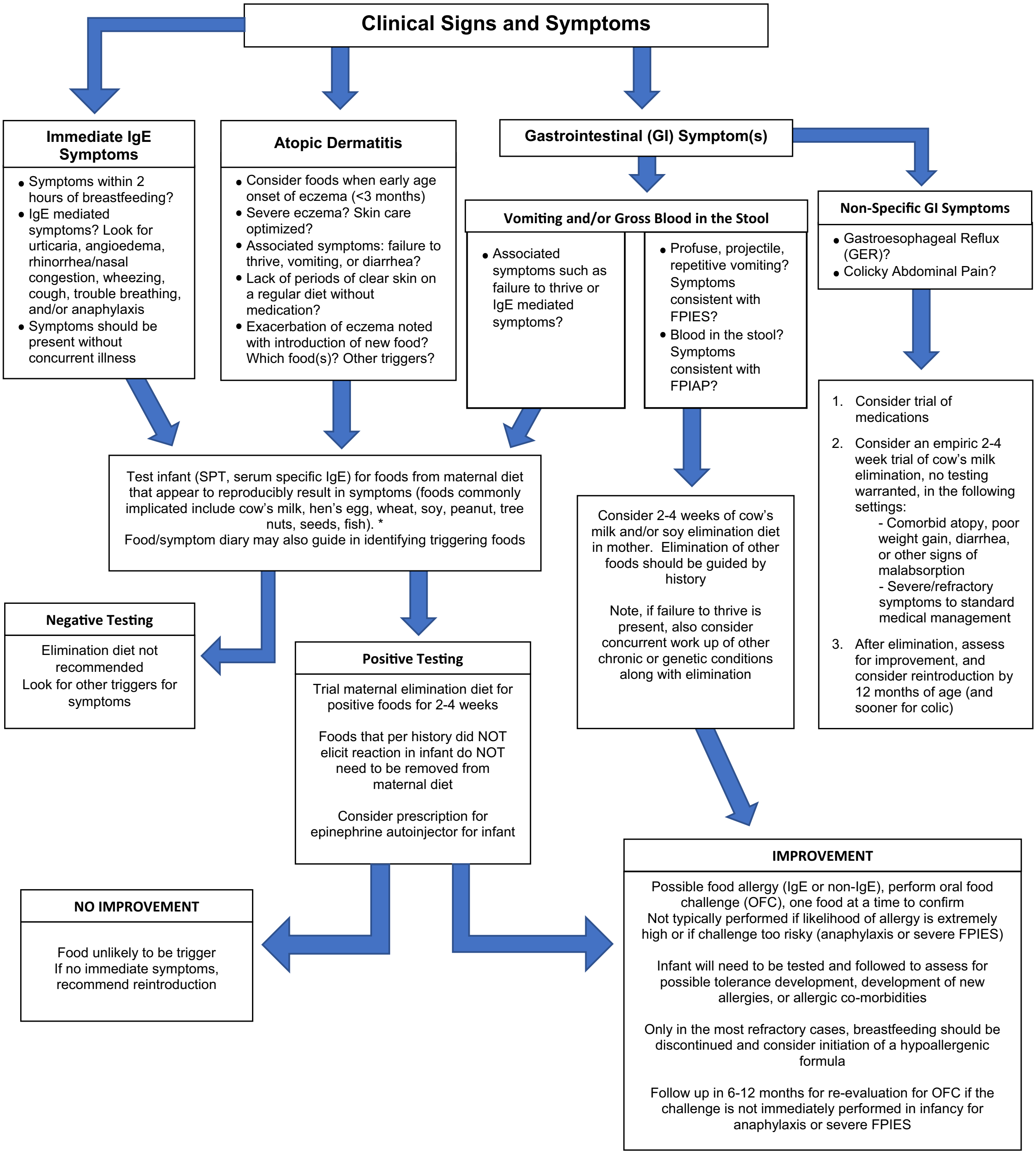

FIGURE 2. Clinical approach and management of suspected food allergy in a breastfeeding Infant. Suggested algorithmic approach to breast-fed infants who present with possible concern for food allergy. Begin with assessment of symptoms the infant is presenting with. If answers to questions in the first row of boxes suggest food as a possible trigger, one should proceed down the arrows to evaluate and manage infants using methods provided. Algorithm is based on related literature with few randomized, controlled data as well as clinical experience. *GI*, Gastrointestinal; *OFC*, oral food challenge. *Advise against panel testing per National Institute of Allergy and Infectious Diseases guidelines.

Infant presenting with atopic dermatitis

As noted above, among children with moderate to severe eczema, about one-third is attributable to food allergy.[7] An earlier onset of and increased severity of eczema increase the risk of food allergy,[52,53] especially to CM, egg, and peanut. There are often associated symptoms such as failure to thrive, vomiting, and diarrhea in infants who have eczema-exacerbated food allergy.[54] Recognizing these statistics can help guide the need for additional workup in an infant presenting with eczema. The first step should be to categorize eczema as mild, moderate, or severe.[55]

TABLE II. Nutrients at risk based on eliminated food

Food eliminated	Nutrients impacted
Milk	Calcium, vitamin D, iodine, riboflavin, phosphorus
Egg	Choline, selenium
Wheat	Thiamin, niacin, riboflavin, folate, iron
Soy	Calcium, iron, magnesium, phosphorus, potassium, niacin, and thiamin
Nuts and seeds	Fats, vitamin E, magnesium
Fish	Long-chain polyunsaturated fatty acids (found in fatty fish)

TABLE III. Alternative sources for nutrients

Nutrient	Alternative dietary sources
Calcium	Fortified plant-based alternative beverages, calcium tofu
Vitamin A	Milk and dairy products, dark green leafy vegetables, deep orange fruits and vegetables, fortified plant-based alternative beverages
Vitamin B_1 (thiamin)	Enriched or fortified grains or whole grains
Vitamin B_2 (riboflavin)	Milk, fortified grains
Vitamin B_{12}	Meat, fish, poultry, eggs, fortified plant-based alternative beverages
Choline	Milk, eggs, peanuts, soy, chicken, beef, liver
Selenium	Brazil nuts, fish, chicken, beef, egg, oats, breads, and cereals
Iodine	Seaweed, fish, iodized salt, milk
Long-chain polyunsaturated fatty acids	Docosahexaenoic acid and eicosapentaenoic acid—fatty fish and fish oils, marine algae oils Alpha-linolenic acid—walnuts, chia, flax, and hemp seeds

A thorough history is required regarding diet and skin care. Food allergies are unlikely to be a trigger if the patient has periods of clear skin on a regular diet without medication.[54] Food allergy is more likely to be a trigger if the skin worsening is temporally related to the introduction of a new food.[56] A food and symptom diary may be helpful to identify problematic foods. It is important to exclude infection (viral, bacterial), irritants, emotional stress, and nonadherence to skin care as other nonfood triggers for eczema. Adherence to skin care is required before performing testing to foods, and adherence rates with topical steroids are reportedly as low as 32%.[57] Furthermore, reiteration of skin care has a benefit in that parental concerns regarding food allergy decrease when the skin is better controlled.[58] Skin care should always be optimized before workup of food allergies, except in the setting of severe uncontrolled eczema or failure to thrive.[8]

When food allergens are suspected to be triggering atopic dermatitis, especially in the setting of moderate to severe atopic dermatitis despite optimal skin care, workup is necessary.[59] Only 35% to 50% of parent-reported allergy is confirmed with a double-blind placebo-controlled food challenge[10,60]; therefore, corroborating history with other objective data, including SPT, serum specific IgE testing, and sometimes open food challenges, may be necessary.[10] Although foods to be tested are dictated by history, testing for commonly implicated food allergens in the population should be considered.[10] Serum specific IgE testing can be complicated by elevated total IgE, which can make it difficult to interpret which results are clinically significant. Once the decision to eliminate foods in mother's diet is pursued on the basis of history and testing, it is crucial to have a period of reintroduction, because spontaneous resolution of eczema can occur. In a small study, although 17 of 37 (46%) infants with atopic dermatitis had "improvement" of eczema with maternal CM and egg elimination, 11 of these 17 had no recurrence of eczema even after reintroduction of these foods.[13] This illustrates the importance of performing a trial of reintroduction after avoidance of an implicated food to assess for worsening of symptoms and therefore confirming that the food is indeed contributing to symptoms. This can help prevent unnecessary dietary restrictions. Given the nuances of a workup of such infants, a stepwise approach is recommended (see Figure 2).

Infant with gastrointestinal manifestations

As noted above, there are a number of non−IgE-mediated reactions, largely gastrointestinal symptoms, that may present as food allergy in breastfed infants, thought to be cell-mediated.[61] In addition to more well-described non−IgE-mediated food allergy disorders such as FPIAP, FPIES, and gastrointestinal eosinophilic syndromes and enteropathies, there are also nonspecific gastrointestinal presentations in infants such as gastroesophageal reflux (GER), constipation, and colic to consider.[62] A summary of guidelines and our recommendations for the management of specific gastrointestinal symptoms is described below.

Gross blood in the stool. The differential of gross blood in the stool should include constipation with anal fissures, reflux esophagitis, inflammatory bowel disease, intussusception, Meckel's diverticulum, vascular malformations, vitamin K deficiency, infectious colitis, Hirschsprung disease, necrotizing enterocolitis, malignancy, and of course, FPIAP. These diagnoses should all be entertained and ruled out on the basis of clinical context. As described above, FPIAP manifests as a healthy baby with gross visualization of specks or streaks of blood mixed with mucus in the stool and is a clinical diagnosis.[62,63] It is important to note that microscopic blood, or positive stool guaiac test results, generally do not warrant further workup for food allergy because it is often unrelated to allergic sensitivity.[64] Furthermore, guaiac tests are generally felt to be overly sensitive, with a high rate of false positives and should be considered as a part of workup in only unique circumstances, such as high concern for severe chronic inflammatory diseases or unexplained failure to thrive.[65-68]

Given the clear role of maternal dietary antigens as triggers in breastfed infants with FPIAP, the mainstay of treatment is strict maternal avoidance of the offending food or foods with continued breastfeeding.[18] Dietary elimination typically begins with exclusion of CM by the mother, given the predominance of this trigger; however, additional dietary restriction may be necessary depending on symptoms.[69] Often, soy is also limited, though soy lecithin and oil are generally tolerated. Skin testing or serum specific IgE testing is not warranted for diagnosis or before introduction into mother's or infant's diet. Most infants show symptom improvement after 72 to 96 hours of dietary avoidance; however, some may take up to a month to demonstrate improvement.[20,70] In 20% of infants, symptoms resolve without maternal dietary changes.[16] In a study evaluating the prognosis of this condition using oral feeding challenges, by 1 year of age more than 50% of infants tolerated their food trigger, and by 3 years, nearly all children had outgrown the allergy.[23] The Academy of Breastfeeding Medicine's protocol provides in-depth information on the management of FPIAP in exclusively breastfed infants, and offers specific guidance based on severity of symptoms and response to dietary changes.[69] We advise against guaiac stool tests to assess for resolution of blood in the stool, and gross examination is sufficient.

Vomiting. Vomiting and "frequent spit up" can be difficult to distinguish, and clear history should be obtained from caregivers. Timing and severity of symptoms can help elucidate an underlying diagnosis. Specifically, delayed onset of repetitive vomiting 1 to 4 hours after a feed, ill-appearance, and lethargy would be suggestive of FPIES, which has been described in exclusively breastfed infants as noted above. Published guidelines for the diagnosis and management of FPIES provide clear recommendations.[71] Per the consensus guidelines, skin testing or serum specific IgE testing can be considered in patients with certain comorbid conditions, specifically with concern for atypical FPIES or IgE-mediated features. Testing should also be considered in patients with FPIES at follow-up visits, as dictated by interval history, because 2% to 20% might eventually have positive test responses to the suspect FPIES-related food(s) and 20% to 40% will have positive test responses to other common food allergens. Atopy patch tests to foods are not currently recommended at any stage.

Regarding EoE, which, as noted above, often presents with vomiting in the younger population, there have been no reports to date of this diagnosis in exclusively breastfed infants. Endoscopy should be saved for individuals who have refractory vomiting and additional symptoms such as dysphagia or failure to thrive. Often, only switching from hypoallergenic formula to CM and expansion of other common allergens to the diet will indicate whether EoE should be further considered.

Reflux. Reflux is common in infants, thought to be physiologic due to the immaturity of the lower esophageal sphincter and chronic supine positioning of infants. Hegar et al[72] prospectively studied infants and showed that in the first 5 months of life, 50% to 72% of infants have episodes of regurgitation at least once per day. CM elimination can improve reflux in a minority of infants, described in as many as 16% to 42% of infants with GER.[73-75] One study suggests that infants who have more severe GER are more likely to have attributable CMA,[76] though in the study, pH monitoring did not show a significant increase in the reflux index during CM challenge. There are no recent studies, and many of these studies are small and have limitations, but are helpful in giving an indication that in clinical practice, a trial elimination diet may be warranted. In most cases where CM avoidance has resulted in improvement of reflux, additional symptoms such as atopic dermatitis, malabsorption, poor weight gain, and diarrhea are described.[62,77,78] Therefore, unless there is presence of comorbid conditions or symptoms are refractory to medical management, given the limitations of the studies reported, we would not recommend CM elimination in the diet of mothers with an infant with reflux[62]; however, a trial of antireflux medications should be considered if symptoms are significant. If after failure of medical therapy a trial of CM elimination is recommended in the maternal diet, consideration should be made for introduction into the maternal/infant's diet at age 12 months, because most reflux does resolve by this time period, even in the presence of associated milk protein allergy.[72,79] IgE testing by SPT or in the serum is not generally indicated. Association of GER disease with other foods besides CM has not been described.

Colicky abdominal pain. The evidence of colic being related to CMA has been reported,[80,81] especially in the gastrointestinal literature, but not consistently. Some clinical trials have shown a treatment benefit of maternal elimination diets in infants with colic,[82,83] though the studies have been small and outcome measurements are largely subjective. The most commonly implicated trigger is CM, and in severe colic, soy formula has actually been reportedly protective.[84] Only 1 study suggested elimination of more than dairy in the maternal diet.[83] The meta-analysis by Lucassen et al[82] recommends that a 1-week trial of maternal CM elimination is warranted, along with reduced stimulation, to assess for improvement in colic symptoms.[82] More recently, this has been corroborated by other reviews that suggest that a hypoallergenic maternal diet may be helpful in colicky infants, especially in the presence of other comorbidities such as bloody stools, vomiting, or atopic dermatitis.[85-87] These studies also acknowledged that better-controlled trials are needed to fully understand the effect of these diets. In all studies, elimination diets were not dictated IgE testing. On the basis of these studies, and recognizing that improvement symptomology is largely subjective, we recommend that it may be appropriate to recommend empiric, short-term elimination of CM in the maternal diet for infants with severe colic, and likely is more strongly warranted in the setting of comorbid atopy in the infant. Also, GER *disease* should be ruled out in these infants because excessive crying may also be due to untreated reflux.

Constipation. In infancy, no consistent association has been made regarding improvement in constipation with maternal elimination of CM. In fact, a more recent study showed in 185 infants and toddlers younger than 2 years (18 of which who were exclusively breastfed had constipation), only 2 infants (1% of total group) showed evidence for constipation related to CMA.[88] With this limited data, we would recommend against elimination diets in exclusively breastfed infants presenting with isolated constipation.

In summary, there are clear IgE- and non−IgE-mediated food allergies that are well described in breastfed infants and there are data to consider for management of maternal diet in these infants. However, for nonspecific gastrointestinal symptoms including reflux and colic, GER disease is excluded first, and IgE testing by SPT or serum specific IgE testing to foods is rarely warranted in the setting of gastrointestinal symptoms. National Institute of Allergy and Infectious Diseases guidelines advise against panel testing in food allergy, and history should guide testing for food allergens, although rarely indicated in gastrointestinal symptoms except in the setting of IgE-mediated immediate symptoms or severe eczema.[89] A 2- to 4-week trial, usually of CM, could be considered. Unique circumstances that may lead a clinician to move straight to potential elimination of foods should be in the setting of failure to thrive, severe uncontrolled atopic dermatitis, or an otherwise unstable infant (see Figure 2).

NUTRITIONAL CONSIDERATIONS

Infants with food allergies must avoid the food to which they are allergic and in a subset of infants, this may include antigen presentation through breastmilk. Breastfeeding is the global standard for infant feeding and in most cases, maternal avoidance diets should be the first step to manage clinical reactions to antigens in human milk rather than discontinuing breastfeeding in favor of a hypoallergenic formula. However, the nutrient composition of human milk is dynamic and dependent on many factors not the least of which is maternal health and diet.

By no means does the maternal diet need to be perfect to nourish an infant well.[90] The maternal diet does affect nutrients presented through breastfeeding, and the elimination of foods of significant nutritional value such as milk, egg, wheat, soy, fish, and nuts and eliminating multiple foods requires dietary planning to support the nutritional viability of breastmilk. Certain nutrients, such as calcium, iron, zinc, copper, and folate, are not highly dependent on maternal diet. Other nutrients such as thiamin, riboflavin, vitamin B_6, vitamin B_{12}, choline, vitamin A, vitamin D, selenium, and iodine as well as the types of fatty acids such as docosahexaenoic acid present in human milk are dependent on maternal diet and therefore require some attention[90-92] (see Table II).

Although fat content as a percentage of macronutrients in human milk remains stable, the types of fatty acids present in human milk are a reflection of the maternal diet and maternal stores.[93] Bzikowska-Jura et al[94] measured maternal dietary omega-3 fatty acid intake via food frequency questionnaire and 3-day food diary and found a significant positive correlation between habitual fatty fish consumption and docosahexaenoic acid concentration in human milk. Eicosapentaenoic and docosahexaenoic acids are important polyunsaturated fatty acids that play a key role in growth, and cognitive and motor systems development. So important are these nutrients that the Food and Drug Administration recommends that children and women who are pregnant or breastfeeding should consume 2 to 3 servings per week of lower mercury fish. Fatty fish offers high-quality protein, micronutrients (vitamins B_{12} and D, selenium, iodine) and long-chain polyunsaturated omega-3 fatty acids. Mothers avoiding fish may not have good sources of these fatty acids; hence, the infant's diet via human milk may also be inadequate. Other dietary sources of omega-3 fatty acids include certain nuts (eg, walnuts) and seeds (eg, flax seed, hemp seed, and chia seed) in the form of alpha- linolenic acid; however, alpha-linolenic acid converts with low efficiency, for example, less than 10% to docosahexaenoic acid and eicosapentaenoic acid. If fish must be avoided in the maternal diet, marina algae-based supplements that provide docosahexaenoic acid and eicosapentaenoic acid can help meet maternal dietary needs.

Carotenoids are important for the visual and cognitive development of an infant. A study from Poland evaluated the content of beta-carotenes lycopene, lutein, and zeaxanthin in human milk and its associations with dietary intake from healthy mothers during the first 6 months of lactation and found a positive correlation between carotenoids in human milk and dietary intake.[95] Another study evaluated diet-dependent components of human milk and found usual maternal daily intakes of vitamin A, niacin, and riboflavin to be positively associated with human milk concentrations.[96] Vitamin D content of human milk is typically quite low but can be impacted to supply adequate vitamin D levels to the infant, only at extremely high maternal supplementation (6400 IU/d), which is not typically recommended.[97] Therefore, vitamin D supplementation is the only nutrient recommended to healthy full-term infants who are exclusively breastfed. In a US study of 103 pregnant women, participants were randomly assigned supplementation with 750 mg choline/d from 18 weeks gestation to 45 days postpartum or no supplementation. Human milk concentrations of free choline, betaine, and phosphocholine were significantly higher in those supplemented.[98] In those not receiving supplemental choline, dietary intake was found to be significantly but weakly correlated with human milk phosphatidylcholine. In the United States, low maternal intakes of vitamin B_6 were associated with lower levels in human milk at 14 days postpartum.[99] Severe maternal deficiency of vitamin B_{12}, related to a vegan diet or undiagnosed pernicious anemia, is documented via case studies in exclusively breastfed infants to result in clinical symptoms of infant deficiency including severe malnutrition.[100]

Although we focus here on infant diet, maternal nutritional status is also at risk. Adams et al[101] evaluated nutritional parameters and bone metabolism in 8 breastfeeding mothers on allergen elimination diets (CM, and stepwise as needed egg, wheat, peanut, tree nuts, grains) and 8 controls on an unrestricted maternal diet. After a 2- to 6-month elimination diet, intakes of total calories, protein, and phosphorus were significantly decreased in those on elimination diets compared with those on unrestricted diets. In addition, more pronounced bone turnover was found in those on diets eliminating CM compared with those on unrestricted diets, even though calcium supplementation was sufficient. Previous studies have indicated that bone loss during lactation is reversible after lactation cessation; however, maternal dietary insufficiency and breastfeeding for longer durations have not been sufficiently studied. Ensuring adequate substitutes or supplementation to the maternal diet may improve the nutritional quality of the maternal diet and the nutritional quality of human milk for the allergic infant. See Table III for alternative sources of nutrients.

Children with food allergies are at increased risk of inadequate nutrient intake and poor growth as demonstrated by numerous studies.[102-107] A severely restricted maternal diet can significantly change the nutrition presented to the infant via human milk. Maternal dietary elimination is not without nutrition risk and should be used only when there is evidence of infant food allergy resulting in allergic reactions after breastfeeding from an unrestricted maternal diet.

CONCLUSIONS AND FUTURE IMPLICATIONS

Given the accumulating evidence, reactions to proteins in the maternal diet delivered via human milk may be less rare than once thought; however, most reports are small and come from referral centers, resulting in uncertain prevalence estimates. A thoughtful approach is necessary in evaluation of infants presenting with concern for possible food allergy–triggered symptoms, which may vary from IgE-mediated allergic symptoms, to atopic dermatitis, to severe gastrointestinal symptoms. The goal should be to minimize unnecessary elimination diets and approach should always be guided by the stability and growth of the infant.

Regarding specific conditions, reports are available for infants who have presented with anaphylaxis, hives, atopic eczema, GER, colic/abdominal pain, vomiting, FPIAP, and acute/chronic FPIES, although similar symptoms can also appear without association with food allergies. Case reports or observational studies of infants presenting with EoE were not found, but with the advances to use less-invasive diagnostic tools,[108] diagnosis in this population may be more easily accomplished. Efforts should be taken in future studies to provide detailed information about infants with EoE to inform clinical decision making in the diagnosis and management of this condition. Additional large studies would also be necessary to determine how commonly FPIES symptoms are induced via human milk, and quantification of the concentration of allergenic proteins in human milk samples from mothers of infants with FPIES who do and do not experience reactions after breastfeeding should be carried out. This would allow for a better understanding of whether the dose of dietary protein, the reaction threshold of the infant, or a combination of factors explain why a small subgroup reacts to their mother's milk, whereas most infants with FPIES do not. In general, strength of data we have reported varies given that most of the data come from case reports and small studies that have many limitations, but perhaps provide a stepping stone for possible studies in the future that could be performed in a randomized manner and on a larger scale.

Published data on the presence of food proteins in human milk are limited. There is no commercially available test for detection of food antigens in human milk, and for the published methods, it is not known whether the detected amounts of antigen are sufficient to have biologic function and elicit a reaction. Future research on detection and function of dietary proteins in human milk is needed. Developing multiplexed allergen detection assay for human milk would enable inclusion of human milk allergen surveillance in assessment of the need of maternal elimination diets in clinical care and could be used to characterize the circumstances under which food antigens are transmitted to milk. This would also enable personalized medicine therapies for breastfed infants who are already suffering from food allergies. Knowing whether a given mother secretes significant levels of food antigens in her milk would eliminate unnecessary maternal elimination diets, which can pose hazards to both the mother and her infant and result in premature discontinuation of breastfeeding. Even more helpful would be to be able to set a threshold on safe levels of food antigens in human milk given to food-allergic infants on the basis of biologic activity of antigens in human milk. Such assays could also be used in future and existing allergy cohorts in estimating infant's exposure to multiple food antigens via human milk and the relation of that exposure to allergy development. In addition, assays that address the biologic activity of food proteins in human milk would be beneficial.

Here, we have provided a novel, initial proposal for an algorithm to aid in the diagnosis and management of exclusively breastfed infants presenting with concern for food allergy. Further studies and consensus guidelines would greatly improve the clinician's ability to manage these often-nuanced cases.

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